

Git

Exercise 1: Initialize Git Repository and Configure

```
MINGW64:/c/Users/laksh/OneDrive/Desktop

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ git version
git version 2.51.1.windows.1

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ git config --global user.name "Bhavya-Tadepalli"

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ git config --global user.email "lakshmitadepalli761@gmail.com"

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ git config --global --list
core.autocrlf
user.email=lakshmitadepalli761@gmail.com
user.name=Bhavya-Tadepalli

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ |
```

```
laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git commit -m "1st commit gitignore"
[master (root-commit) 970a99e] 1st commit gitignore
1 file changed, 2 insertions(+)
create mode 100644 .gitignore

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ |
```

```
laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ cd gitdemo

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ pwd
/c/Users/laksh/OneDrive/Desktop/gitdemo

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ mkdir log

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ ls
log/

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ touch error.log

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ echo "This is a log file" > error.log

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo
$ ls
error.log  log/
```

```
laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git commit -m "1st commit gitignore"
[master (root-commit) 970a99e] 1st commit gitignore
1 file changed, 2 insertions(+)
create mode 100644 .gitignore

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git status
On branch master
nothing to commit, working tree clean

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ |
```

Exercise 2: Branch Creation and Merging

```

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitNewBranch)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitNewBranch)
$ git commit -m "after adding file1"
[GitNewBranch 3af3d66] after adding file1
1 file changed, 1 insertion(+)
create mode 100644 file1.txt

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git diff master GitNewBranch
diff --git a/file1.txt b/file1.txt
new file mode 100644
index 0000000..d9209a7
--- /dev/null
+++ b/file1.txt
@@ -0,0 +1 @@
+This is my first brach.

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git merge GitNewBranch
Updating 970a99e..3af3d66
Fast-forward
 file1.txt | 1 +
1 file changed, 1 insertion(+)
create mode 100644 file1.txt

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git log
commit 3af3d6606141509175d5cd96aade1b048302110e (HEAD -> master, GitNewBranch)
Author: Bhavya-Tadepalli <lakshmitadepalli761@gmail.com>
Date: Sat Jul 25 12:25:57 2026 +0530

    after adding file1

commit 970a99e69f494fdeb96610d7b06f4e445022c0ed
Author: Bhavya-Tadepalli <lakshmitadepalli761@gmail.com>
Date: Sat Jul 25 12:20:55 2026 +0530

    1st commit gitignore

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 3af3d66).

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git status

```

Exercise 3: Merge Conflict Resolution

```
MINGW64:/c/Users/laksh/OneDrive/Desktop/gitdemo

laksh@Bhavya MINGW64 ~/OneDrive/Desktop
$ cd gitdemo

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git checkout master
Already on 'master'

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git branch GitWork

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git checkout GitWork
Switched to branch 'GitWork'

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ touch hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ echo "<message> helloo from gitwork branch</message>" > hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ cat hello.xml
<message> helloo from gitwork branch</message>

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ git commit -m "Added hello.xml in git work branch"
[GitWork 833fc98] Added hello.xml in git work branch
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (GitWork)
$ git checkout master
Switched to branch 'master'

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ touch hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ echo "<message>hello from master branch</message>" > hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ cat hello.xml
<message>hello from master branch</message>
```

MINGW64:/c/Users/laksh/OneDrive/Desktop/gitdemo

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

\$ git add hello.xml

warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

\$ git commit -m "added hello in master"

[master e6539fd] added hello in master

1 file changed, 1 insertion(+)

create mode 100644 hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

\$ git log --oneline --graph --decorate --all

* e6539fd (HEAD -> master) added hello in master

| * 833fc98 (GitWork) Added hello.xml in git work branch

|/

* 3af3d66 after adding file1

* 970a99e 1st commit gitignore

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

\$ git merge GitWork

Auto-merging hello.xml

CONFLICT (add/add): Merge conflict in hello.xml

Automatic merge failed; fix conflicts and then commit the result.

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ cat hello.xml

<<<<<< HEAD

<message>hello from master branch</message>

=====

<message> helloo from gitwork branch</message>

>>>>>> GitWork

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ echo "<message>Hello from Master Branch</message>" > hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ echo "<message>Hello from GitWork Branch</message>" >> hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ cat hello.xml

<message>Hello from Master Branch</message>

<message>Hello from GitWork Branch</message>

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ git add hello.xml

warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master|MERGING)

\$ git commit -m "Resolved merge conflict in hello.xml"

[master cfde7c9] Resolved merge conflict in hello.xml

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

\$ git status

On branch master

nothing to commit, working tree clean

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)

Exercise 4: Remote Repository (GitHub)

```
laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git remote set-url origin https://github.com/Bhavya-Tadepalli/gitdemo.git

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git remote -v
origin https://github.com/Bhavya-Tadepalli/gitdemo.git (fetch)
origin https://github.com/Bhavya-Tadepalli/gitdemo.git (push)

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git push -u origin master
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (15/15), 1.27 KiB | 432.00 KiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/Bhavya-Tadepalli/gitdemo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
$ git pull origin master
From https://github.com/Bhavya-Tadepalli/gitdemo
 * branch            master      -> FETCH_HEAD
Already up to date.

laksh@Bhavya MINGW64 ~/OneDrive/Desktop/gitdemo (master)
```

The screenshot shows the GitHub web interface for the repository `gitdemo` by user `Bhavya-Tadepalli`. The repository is public and has 1 branch and 0 tags. The main branch is `master`. The commit history shows 5 commits. The most recent commit, `cfde7c9`, was made 13 minutes ago and resolved a merge conflict in `hello.xml`. The commit message is "Resolved merge conflict in hello.xml". The commit history table lists the following files and their commit messages:

File	Commit Message	Time
<code>.gitignore</code>	1st commit gitignore	51 minutes ago
<code>file1.txt</code>	after adding file1	46 minutes ago
<code>hello.xml</code>	Resolved merge conflict in hello.xml	13 minutes ago

